

- 6 Water absorbed by plant roots travels by different pathways from root hairs to the xylem.

Fig. 6.1 shows these pathways in the root of *Ranunculus acris*.

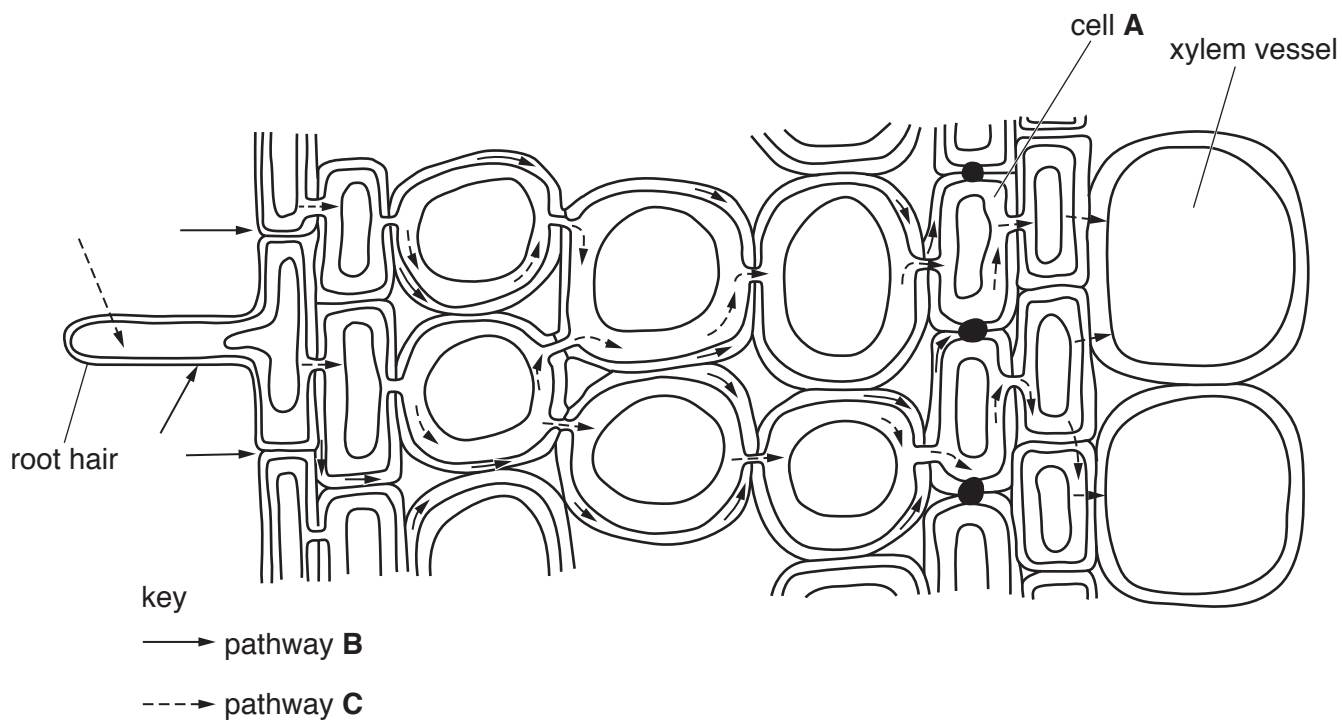


Fig. 6.1

- (a) Name cell **A** and pathway **B** as shown in Fig. 6.1.

cell **A**

pathway **B**

[2]

- (b)** Transpiration occurring at the leaves is mainly responsible for movement of water across the root of *R. acris*.

Explain how transpiration is responsible for the movement of water across the root as shown in Fig. 6.1.

.....[5]

- (c)** Explain why the movement of water in pathway **C** is slower than in pathway **B**.

[2]

- (d) Ions are taken up by root hair cells using active transport and facilitated diffusion.

Describe **two** ways in which facilitated diffusion differs from active transport.

1

.....

.....

2

.....

.....

[2]

[Total: 11]

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